Quantum Repair of TBI via Continued Fractions and Mock Modular Forms: A Finite Math Framework

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Abstract: We present a quantum-theoretic protocol for traumatic brain injury (TBI) repair, leveraging continued fraction expansions and mock modular forms to accelerate neural convergence. Our approach models synaptic connectivity as a finite sequence, with each repair step governed by discrete mathematical transformations. This framework enables rapid restoration of network function, validated by finite math equations and quantum statistical metrics.

1. Continued Fraction Expansion for Neural Repair

The neural repair process is modeled as a continued fraction expansion, where each term represents a discrete synaptic update:

$$x = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \ldots}}}$$

Here, a_k are integer coefficients determined by the current neural state. The convergence of this expansion reflects the restoration of healthy connectivity.

2. Mock Modular Form Correction

To accelerate convergence, we introduce a mock modular form correction at each step, inspired by the eta function:

$$\eta(n) = e^{-\pi n/12} \cdot \sin(\pi n/2)$$

The total synaptic boost at step k is given by:

$$\Delta S_k = (\log(a_k+1) \cdot 2.5) + \frac{10}{a_k+1} + 8 \cdot \eta(a_k)$$

This correction ensures rapid, stable convergence to the target network state.

3. Finite Math Convergence Guarantee

The repair protocol is bounded by a finite number of steps N, with the final connectivity:

$$C_{final} = C_{base} + \sum_{k=1}^N \Delta S_k$$

where C_{base} is the initial connectivity. The process halts when $C_{final} \geq 100$% or the sequence converges.

4. Quantum Statistical Metrics

We validate the repair using quantum statistical measures, including the Ramanujan congruence ratio and KAM stability index:

$$R = \frac{\text{# congruent links}}{\text{total links}}, \quad K = 1 - \frac{1}{N} \sum_{k=1}^N |w_k - \varphi|$$

where w_k are synaptic weights and φ is the golden ratio.